Matrices & Vectors

ANSWER KEY

Section 1: Matrix Inversion & Systems

1 What is the inverse of $A = [[4, 7], [1, 2]]$?

a. $[[2, -7], [-1, 4]]$ ✓

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

$\det(A) = 4(2) - 7(1) = 8 - 7 = 1$. Inverse is $\frac{1}{1}[[d, -b], [-c, a]] = [[2, -7], [-1, 4]]$.

2 For what value of k does the matrix $[[2, k], [4, 6]]$ fail to be invertible?

a. 3 ✓

b. 6

c. 12

d. 0

Non-invertible when determinant is zero: $2(6) - 4k = 0 \Rightarrow 12 = 4k \Rightarrow k = 3$.

3 If $AX = B$, where A is invertible, which expression gives the solution column vector X?

a. $A^{-1}B$ ✓

b. BA^{-1}

c. B/A

d. AB^{-1}

Multiply on the left by A^{-1}: $A^{-1}AX = A^{-1}B \Rightarrow X = A^{-1}B$.

Section 2: Determinants & Properties

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4 If A and B are 3×3 matrices with $\det(A) = 4$ and $\det(B) = -2$, what is $\det(AB)$?

a. -8 ✓

b. 8

c. -2

d. 2

By the multiplicative property of determinants: $\det(AB) = \det(A) \times \det(B) = 4 \times (-2) = -8$.

5 If M is a 2×2 matrix with $\det(M) = 6$, what is $\det(M^{-1})$?

a. $\frac{1}{6}$ ✓

b. -6

c. 6

d. 36

$\det(M^{-1}) = \frac{1}{\det(M)} = \frac{1}{6}$.

6 What happens to the determinant of a matrix if two of its rows are interchanged?

a. **Its sign is reversed** ✓

b. It doubles

c. It remains unchanged

d. It becomes zero

Interchanging any two rows of a square matrix multiplies its determinant by -1.

Section 3: Dot Product & Projections

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7 Find the angle between $\mathbf{u} = \langle 2, 2 \rangle$ and $\mathbf{v} = \langle 0, 3 \rangle$.

a. 45° ✓

b. 30°

c. 60°

d. 90°

$\cos\theta = \frac{2(0)+2(3)}{\sqrt{8}\times 3} = \frac{6}{3\sqrt{8}} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$.

8 What is the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$?

a. $\langle 3, 0 \rangle$ ✓

b. $\langle 4, 0 \rangle$

c. $\langle 0, 4 \rangle$

d. $\langle 1, 0 \rangle$

$\text{proj}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v} = \frac{3(1)+4(0)}{1^2}\langle 1, 0 \rangle = 3\langle 1, 0 \rangle = \langle 3, 0 \rangle$.

9 If vectors $\mathbf{a} = \langle 2, x \rangle$ and $\mathbf{b} = \langle 6, -4 \rangle$ are perpendicular, what is x?

a. 3 ✓

b. -3

c. 4

d. -4

Dot product must be zero: $2(6) + x(-4) = 0 \Rightarrow 12 - 4x = 0 \Rightarrow x = 3$.

Section 4: 3D Vectors & Cross Products

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10 Compute the cross product $\langle 1, 0, 0\rangle \times \langle 0, 1, 0\rangle$.

a. $\langle 0, 0, 1\rangle$ ✓

b. $\langle 0, 0, -1\rangle$

c. $\langle 1, 1, 0\rangle$

d. $\mathbf{0}$

$\mathbf{i} \times \mathbf{j} = \mathbf{k} = \langle 0, 0, 1\rangle$.

11 What is the volume of the parallelepiped determined by vectors $\mathbf{u} = \langle 1, 0, 0\rangle$, $\mathbf{v} = \langle 0, 2, 0\rangle$, and $\mathbf{w} = \langle 0, 0, 3\rangle$?

a. 6 ✓

b. 12

c. 3

d. 1

Volume is absolute value of scalar triple product: $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})| = |1(2 \times 3)| = 6$.

12 What is the unit vector in the direction of $\mathbf{v} = \langle 3, -6, 2\rangle$?

a. $\langle \frac{3}{7}, -\frac{6}{7}, \frac{2}{7}\rangle$ ✓

b. $\langle \frac{3}{9}, -\frac{6}{9}, \frac{2}{9}\rangle$

c. $\langle 1, -2, \frac{2}{3}\rangle$

d. $\langle \frac{3}{49}, -\frac{6}{49}, \frac{2}{49}\rangle$

Magnitude $\|\mathbf{v}\| = \sqrt{3^2 + (-6)^2 + 2^2} = \sqrt{9 + 36 + 4} = \sqrt{49} = 7$. Unit vector is $\frac{\mathbf{v}}{7}$.

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